# EAB 770 - Chapter 2

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# 1 Chapter 2: $2 \times 2$ tables

## 1.1 Outline

- Contingency Tables
- Chi-Square Statistics

- Exact Tests
- Difference in Proportions
- Odds Ratio and Relative Risk
- Sensitivity and Specificity
- McNemar's Test
- Incidence Densities
- Power Analysis

# 2 Contingency Tables

## 2.1 2x2 Contingency Tables

- Summarizes categorical data
  - Response variable has two levels
  - Exploratory variable has two levels
- Example:
  - Treatment/Placebo in RCT
  - Case/Control in a retrospective study

| Risk\Case | Case | Control | Total |
|---|---|---|---|
| Present | | | |
| Absent | | | |
| Total | | | |

## 2.2 Contingency Tables

We wish to determine whether there is an association between the row and column variables using the frequencies observed in each category.

> **Note**
>
> Is the risk variable associated with the prevalence rate?

Recall: Intro to Statistics

- Chi-square tests of homogeneity $H_0$: The two populations (rows) have the same distribution of the column variable
- Chi-square tests of independence $H_0$: The row and column variables are independent of each other.

## 2.3 Contingency Tables: Convention

The indices of contingency tables denote the row and column of each number. For a total sample size of $n$,

- $n_{xy}$ denotes the number at row $x$ and column $y$, also referred to as a joint frequency.
- Seeing a (+) sign indicates that it is a marginal sum, also referred to as marginal frequencies.
  - $n_{x+}$ denotes the sum of all columns in row $x$.
  - $n_{+y}$ denotes the sum of all rows in column $y$.
- The placement of the + sign denotes the margin in which it was summed.

| Row\Col | $C_1$ | $C_2$ | Total |
|---|---|---|---|
| $R_1$ | $n_{11}$ | $n_{12}$ | $n_{1+}$ |
| $R_2$ | $n_{21}$ | $n_{22}$ | $n_{2+}$ |
| Total | $n_{+1}$ | $n_{+2}$ | $n$ |

## 2.4 SAS Example: Contingency Table

```
data respire;
length treat $7. outcome $1. ;
input treat $ outcome $ count;
datalines;
placebo f 16
placebo u 48
test f 40
test u 20
;
run;
```

```
proc freq data=respire order=data;
weight count;
tables treat*outcome;
run;
```

- `proc freq` calls the `FREQ` procedure in `SAS`.
  - Not defining `data=respire` means `SAS` will automatically use the most recently run data set.
- `order=data` means we are creating the rows and columns based on how the data was inputted, not alphabetical order.
- `weight` tells `SAS` the number of counts for each count
- `tables` creates the contingency table with row, column, and total percentages.
- `run;` tells `SAS` to run the code. **This is important!**

## 2.5 Contingency Tables in `SAS`- Long Format

When the data is not aggregated, the `FREQ` procedure can still be used to create contingency tables. We can use the import data procedure in SAS ODA to create `work.spore` from `Spore.csv` uploaded on Canvas.

```
proc freq order=data data=severe;
weight count;
tables treat*outcome / chisq nocol;
exact or;
run;
```

## 2.6 Contingency Tables in `R`

To create your own contingency table, you can use the `data.frame()` and `xtabs()` functions.

```
count_data <- data.frame(Row = c("A","A","B","B"),
                         Column = c("X","Y","X","Y"),
                         Count = c(2,4,6,8))
count_data
```

```
  Row Column Count
1   A      X     2
2   A      Y     4
3   B      X     6
4   B      Y     8
```

```
xtabs(Count~Row+Column,data=count_data)
```

```
   Column
Row X Y
  A 2 4
  B 6 8
```

You can also use the `expand.grid()` function to create combinations of variable levels.

```
count_data <- expand.grid(Row = factor(c("A","B"), levels=c("A","B")),
                          Column = factor(c("X","Y"), levels=c("X","Y")))

count_data$Count = c(2,4,6,8)

count_data
```

```
  Row Column Count
1   A      X     2
2   B      X     4
3   A      Y     6
4   B      Y     8
```

```
xtabs(Count~Row+Column,data=count_data)
```

```
   Column
Row X Y
  A 2 6
  B 4 8
```

> **Note**
>
> `ordered` can also be used to account for ordering in the row and column variables.

## 2.7 Contingency Tables in R- Long Format

When the data is not aggregated, the `xtabs` function can still be used to create contingency tables. First, we import `Spore.csv` from Canvas.

```r
spore <- read.csv("Spore.csv",stringsAsFactors = TRUE)
spore
```

```
   ID Treatment         Period Spore
1   1      Test  Pre-Treatment  Full
2   1      Test  Mid-Treatment  Some
3   1      Test Post-Treatment  Some
4   2      Test  Pre-Treatment  Full
5   2      Test  Mid-Treatment  None
6   2      Test Post-Treatment  None
7   3      Test  Pre-Treatment  Full
8   3      Test  Mid-Treatment  Some
9   3      Test Post-Treatment  Some
10  4      Test  Pre-Treatment  Full
11  4      Test  Mid-Treatment  Some
12  4      Test Post-Treatment
13  5      Test  Pre-Treatment  Full
14  5      Test  Mid-Treatment  None
15  5      Test Post-Treatment  None
16  1   Placebo  Pre-Treatment  Full
17  1   Placebo  Mid-Treatment  Full
18  1   Placebo Post-Treatment  Full
19  2   Placebo  Pre-Treatment  Full
20  2   Placebo  Mid-Treatment  Some
21  2   Placebo Post-Treatment  Some
22  3   Placebo  Pre-Treatment  Full
23  3   Placebo  Mid-Treatment  Some
24  3   Placebo Post-Treatment  Some
25  4   Placebo  Pre-Treatment  Full
26  4   Placebo  Mid-Treatment  Some
27  4   Placebo Post-Treatment  Some
28  5   Placebo  Pre-Treatment  Some
29  5   Placebo  Mid-Treatment
30  5   Placebo Post-Treatment
31  6      Test  Pre-Treatment  Full
32  6      Test  Mid-Treatment  Some
33  6      Test Post-Treatment  Some
34  7      Test  Pre-Treatment  Full
35  7      Test  Mid-Treatment  None
36  7      Test Post-Treatment  None
37  8      Test  Pre-Treatment  Full
38  8      Test  Mid-Treatment  Some
```

```
39  8      Test Post-Treatment  Some
40  9      Test  Pre-Treatment  Full
41  9      Test  Mid-Treatment  Some
42  9      Test Post-Treatment  None
43 10      Test  Pre-Treatment  Full
44 10      Test  Mid-Treatment  Some
45 10      Test Post-Treatment  Some
46  6   Placebo  Pre-Treatment  Full
47  6   Placebo  Mid-Treatment  Full
48  6   Placebo Post-Treatment  Full
49  7   Placebo  Pre-Treatment  Full
50  7   Placebo  Mid-Treatment  Full
51  7   Placebo Post-Treatment  Full
52  8   Placebo  Pre-Treatment  Full
53  8   Placebo  Mid-Treatment  Full
54  8   Placebo Post-Treatment  Full
55  9   Placebo  Pre-Treatment  Full
56  9   Placebo  Mid-Treatment  Some
57  9   Placebo Post-Treatment  Some
58 10   Placebo  Pre-Treatment  Full
59 10   Placebo  Mid-Treatment  Some
60 10   Placebo Post-Treatment  Some
61 11      Test  Pre-Treatment  Full
62 11      Test  Mid-Treatment  Some
63 11      Test Post-Treatment  None
64 12      Test  Pre-Treatment  Full
65 12      Test  Mid-Treatment  Some
66 12      Test Post-Treatment  None
67 13      Test  Pre-Treatment  Full
68 13      Test  Mid-Treatment  Some
69 13      Test Post-Treatment  Some
70 14      Test  Pre-Treatment  Full
71 14      Test  Mid-Treatment  None
72 14      Test Post-Treatment  None
73 15      Test  Pre-Treatment  Full
74 15      Test  Mid-Treatment  None
75 15      Test Post-Treatment  None
76 11   Placebo  Pre-Treatment  Full
77 11   Placebo  Mid-Treatment  Some
78 11   Placebo Post-Treatment  Some
79 12   Placebo  Pre-Treatment  Full
80 12   Placebo  Mid-Treatment  Some
81 12   Placebo Post-Treatment  Some
```

```
82 13   Placebo  Pre-Treatment  Full
83 13   Placebo  Mid-Treatment  Some
84 13   Placebo Post-Treatment  Some
85 14   Placebo  Pre-Treatment  Full
86 14   Placebo  Mid-Treatment  Full
87 14   Placebo Post-Treatment  Full
88 15   Placebo  Pre-Treatment  Full
89 15   Placebo  Mid-Treatment  Full
90 15   Placebo Post-Treatment  Full
```

```r
xtabs(~Treatment+Spore,data=spore)
```

```
          Spore
Treatment    Full None Some
  Placebo  2   26    0   17
  Test     1   15   13   16
```

## 2.8 Exercise

Suppose a survey asking sophomores and senior students whether they agree with the statement "I enjoyed my in-person classes this semester." yielded the following results.

Enter the data in SAS using the data step and generate the contingency tables. Name the data set enjoy.

| | Strongly Agree | Agree | Neutral | Disagree | Strongly Disagree |
|---|---|---|---|---|---|
| Sophomore | 24 | 43 | 12 | 25 | 12 |
| Seniors | 19 | 37 | 24 | 32 | 21 |

### 2.8.1 R Code

OR

```r
enjoy <- expand.grid(year=factor(c("Sophomore","Seniors"), levels = c("Sophomore","Seniors"))
                     response = factor(c("SA","A","N","D","SD"),levels = c("SA","A","N","D","
enjoy$Count <- c(24,19,43,37,12,24,25,32,12,21)
xtabs(Count~year+response,data=enjoy)
```

```
           response
year        SA  A  N  D SD
  Sophomore 24 43 12 25 12
  Seniors   19 37 24 32 21
```

### 2.8.2 SAS Code

There are many ways to code the `data` step. Here's one way to do it!

```
data enjoy;
length year $9. response $2.;
input year $ response $ count @@;
datalines;
sophomore SA 24 sophomore A 43
sophomore N 12
sophomore D 25 sophomore SD 12
senior SA 19 senior A 37
senior N 24
senior D 32 senior SD 21
;

proc freq order=data data=enjoy;
weight count;
tables year*response;
run;
```

## 2.9 Independence

If the two categorical variables are independent of each other, they are not associated.

> **Note**
>
> Independence also means that the joint probability $p_{ij}$ can be determined solely on the basis of the marginal probabilities. These can be estimated using sample data.

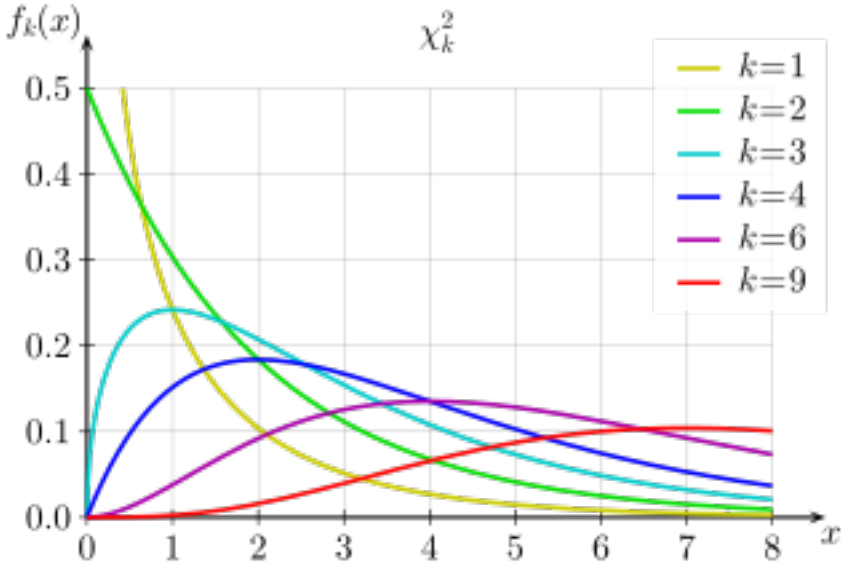
# 3 Chi-square Statistics

## 3.1 Chi-square Statistics

The null hypothesis of no association (independence or homogeneity) between row and column variables imply that the totals are fixed. The probabilities of realization of each cell is determined by the hypergeometric distribution.

## 3.2 Chi-square distribution

- The chi-square distribution, $\chi^2_k$, is defined by the number of degrees of freedom , where 1.
- Properties include:
  - The value of $\chi^2$ can only assume non-negative values.
  - The $\chi^2$ distribution is asymmetric.
  - The sum of two or more independent chi-square variables also follows a chi-square distribution.
  - Mean: $k$, Variance: $2k$




## 3.3 Pearson Chi-square Statistic

The Pearson Chi-square statistic can be written in terms of the joint frequency `n_ij` and the expected frequency `m_ij`.

$$Q_P = \sum_{i=1}^{2}\sum_{j=1}^{2} \frac{(n_{ij} - m_{ij})^2}{m_{ij}}$$

> **Note**
>
> $m_{ij}$ is the expected value of the number of observations in row $i$ and column $j$, $n_{ij}$. The expected frequency $m_{ij}$ can be calculated based on the null hypothesis assumption.
>
> - Independence: $p_{ij} = p_{i+}p_{+j} \rightarrow m_{ij} = \frac{n_{i+}n_{+j}}{n}$
> - Homogeneity: $p_{i+} = n_{i+}/n; p_{+j} = n_{+j}/n \rightarrow m_{ij} = \frac{n_{i+}n_{+j}}{n}$

### 3.4 Mantel-Haenszel Chi-square statistic

Another form of the Chi-square statistic is the Mantel-Haenszel chi-square statistic, $Q$. For $2 \times 2$ contingency tables,

$$Q = \frac{(n_{11} - m_{11})^2}{v_{11}}$$

$$m_{11} = n_{1+}n_{+1}/n; v_{11} = \frac{n_{1+}n_{2+}n_{+1}n_{+2}}{n^2(n-1)}$$

> **Note**
>
> $Q$, also known as the randomization chi-square, is based on the condition that all marginal totals are fixed and the probability distribution of random allocation of units to groups is given by the hypergeometric distribution. $m_{11}$ and $v_{11}$ are the mean and variance of the hypergeometric distribution based on the $2 \times 2$ table, respectively.

### 3.5 Pearson Chi-square vs. Mantel-Haenszel Chi-square

The relationship can also be approximated by the following relation:

$$Q_P = \frac{n}{n-1}Q$$

For large $n$, both the MH chi-square and the Pearson chi-square approach a chi-square distribution with 1 degree of freedom ($\chi_1^2$).

> **Warning**
>
> Rule of thumb for $Q_P$: The expected value for each cell should be at least 5, i.e. $m_{ij} \geq 5$ for all $i, j$

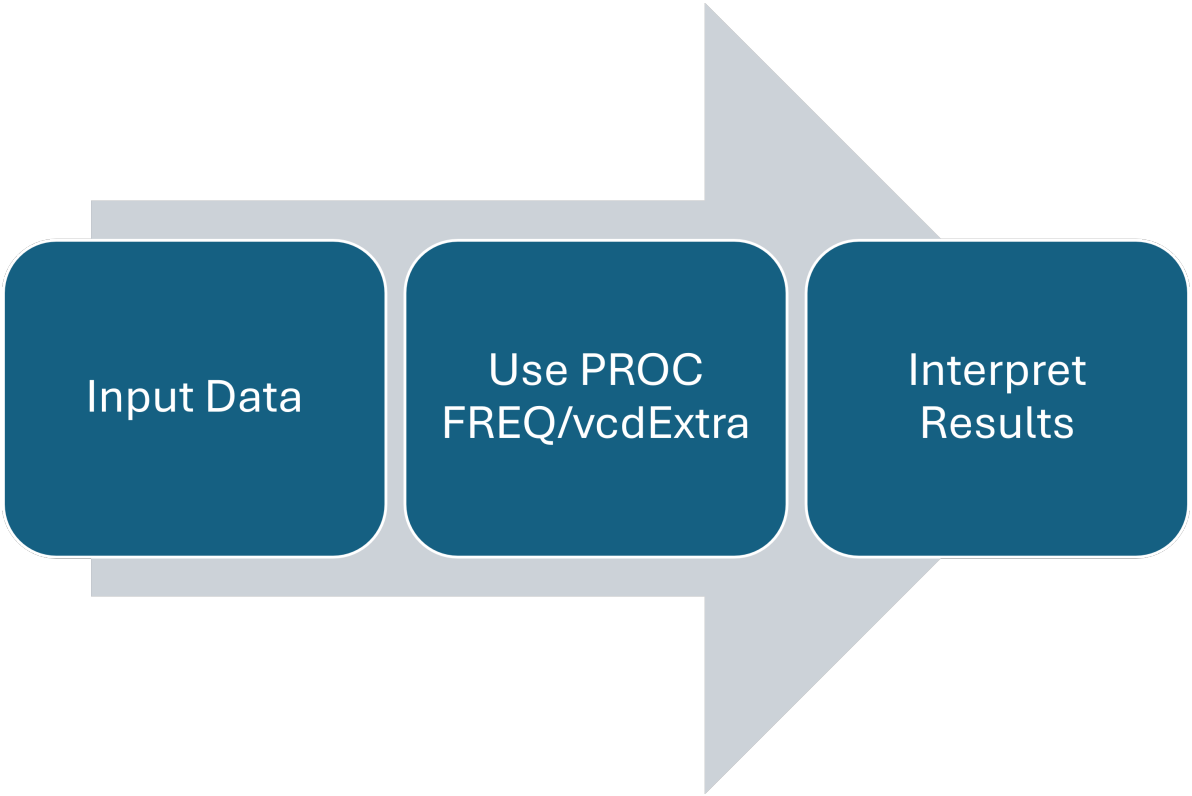
> **Tip**
>
> Because the MH chi-square statistic comes from a hypergeometric distribution, the MH chi-square statistic can be used for small samples using exact calculations of p-values from the hypergeometric distribution. This is not possible with the Pearson chi-square.
> The MH chi-square statistic can also be extended for use in the presence of covariates/stratifying variables.

## 3.6 Hypothesis testing

The null hypothesis of no association states that there is no association between row and column variables. - Assuming this is true and assumptions are satisfied, $Q$ and $Q_P$ follow a $\chi^2_1$ distribution for $2 \times 2$ tables.

> **Note**
>
> The p-values are given by $P(\chi^2 > Q_P)$ and $P(\chi^2 > Q)$ for Pearson and MH Chi-squares, respectively.

### 3.7 Hypothesis Testing: SAS




### 3.8 SAS: FREQ procedure

- The `weight` statement tells SAS if there is an aggregated count variable present in the data set.
- The `tables` statement tells SAS to create a contingency table using the stated variables.
  - Under tables, you can include the `cmh` or the `chisq` option to calculate the randomization and Pearson chi-square values.

Shown below is a sample `SAS` code.

```
proc freq;
weight count;
tables a*b/ cmh chisq;
run;
```

## 3.9 R: vcdExtra

In R, the vcdExtra package is required to perform the Mantel-Haenszel chi-square tests. The function CMHtest() provides the value of the Mantel-Haenszel statistic, while chisq.test() provides the Pearson chi-square statistic.

## 3.10 Example

The table shows the information from a randomized clinical trial that compared an experimental treatment to a placebo treatment for a respiratory disorder.

```
library(tidyverse)
dat <- expand.grid(Treatment=factor(c("Placebo","Test"),levels=c("Placebo","Test")),
                   Outcome = factor(c("Favorable","Unfavorable"),levels=c("Favorable","Unfavo

dat$count = c(16,40,48,20)

xtabs(count ~ .,data=dat)
```

```
         Outcome
Treatment Favorable Unfavorable
  Placebo        16          48
  Test           40          20
```

### 3.10.1 Questions

- What is the Pearson chi-square value?
- What is the Mantel-Haenszel chi-square value?
- Is there strong evidence of an association between treatment and outcome?

### 3.10.2 SAS Code

```
SAScode <- "
data respire;
input treat $ outcome $ count;
datalines;
test f 40
test u 20
placebo f 16
placebo u 48
```

```
;

proc freq data=respire order=data;
weight count;
tables treat*outcome/chisq;
run;
"
```

### 3.10.3 R Code

```
dat <- expand.grid(Treatment=c("Placebo","Test"),
                   Outcome = c("Favorable","Unfavorable")) |>
mutate(count = c(16,40,48,20))

xtabs(count ~ .,data=dat) -> dat_cc
```

Pearson Chi-square (without continuity adjustment)

```
#install.packages("tidyverse")
library(tidyverse)
dat <- expand.grid(Treatment=c("Placebo","Test"),
                   Outcome = c("Favorable","Unfavorable")) |>
mutate(count = c(16,40,48,20))

xtabs(count ~ .,data=dat)
```

```
         Outcome
Treatment Favorable Unfavorable
  Placebo        16          48
  Test           40          20
```

```
chisq.test(dat_cc,correct=F)
```

```
    Pearson's Chi-squared test

data:  dat_cc
X-squared = 21.709, df = 1, p-value = 3.174e-06
```

Randomization Chi-square

```
# install.packages("vcdExtra")
library(vcdExtra)
CMH_CC <-CMHtest(dat_cc)
CMH_CC
```

```
Cochran-Mantel-Haenszel Statistics for Treatment by Outcome

                 AltHypothesis  Chisq Df       Prob
cor        Nonzero correlation 21.534  1 3.4768e-06
rmeans  Row mean scores differ 21.534  1 3.4768e-06
cmeans  Col mean scores differ 21.534  1 3.4768e-06
general    General association 21.534  1 3.4768e-06
```

### 3.10.4 Answers

- What is the Pearson chi-square value? **21.7**
- What is the Mantel-Haenszel chi-square value? **21.53**
- Is there strong evidence of an association between treatment and outcome?
  - $p < 0.0001$; There is strong evidence of an association between the test treatment and the outcome.

## 3.11 Exercise

Suppose we wish to determine if the frequency of falls is related to wheelchair use for Polio survivors. The data is shown in the table.

```
            Wheelchair
Fall         Yes  No
  Fallers     62 121
  Nonfallers  18  32
```

### 3.11.1 Questions

- Write the code to enter the data and analyze using the FREQ procedure.
- What is the Pearson chi-square value?
- What is the Mantel-Haenszel chi-square value?
- Using a significance level of 0.05, do we have sufficient evidence to warrant the conclusion that wheelchair use and falling are associated?

### 3.11.2 SAS code

```
data wheelchair;
input fall $ wheelchair $ count;
datalines;
f yes 62
n yes 18
f no 121
n no 32
;

proc freq data=wheelchair order=data;
weight count;
tables fall*wheelchair/chisq;
run;
```

### 3.11.3 R

Pearson Chi-square (without continuity adjustment)

```
#install.packages("tidyverse")

chisq.test(dat_cc, correct = F)
```

```
    Pearson's Chi-squared test

data:  dat_cc
X-squared = 0.078299, df = 1, p-value = 0.7796
```

Randomization Chi-square

```
# install.packages("vcdExtra")
CMH_CC <-CMHtest(dat_cc)
CMH_CC$table
```

```
             Chisq Df     Prob
cor     0.07796307  1 0.780077
rmeans  0.07796307  1 0.780077
cmeans  0.07796307  1 0.780077
general 0.07796307  1 0.780077
```

# 4 Exact Methods

## 4.1 Fisher's Exact Test

The chi-square distribution assumption for $Q$ and $Q_P$ includes a large sample assumption.

> **Warning**
>
> What if $m_{ij} < 5$?

The **Fisher's Exact Test** can be used to analyze these contingency tables.

## 4.2 Fisher's Exact Test

Fisher's Exact Test is based on the hypergeometric distribution. The test calculates an *exact* p-value, unlike the chi-square test which only uses an approximation.

> **Note**
>
> The test is easy to calculate for $2 \times 2$ and total sample sizes below 50, but computationally expensive as the number increases.

The p-value can be calculated using the hypergeometric distribution with the following parameters: $Hyper(N = n, m = n_{+1}, k = n_{1+})$

## 4.3 Fisher's Exact Test: Hypotheses

Null hypothesis $H_0$: there is no association between the row and column variables

- One sided $H_a$: There is a positive/negative association between the row and column variables. OR One combination of the row and column variable levels is more favored compared to the others.
- Two-sided $H_a$: there is an association between the row and column variables.

> **Note**
>
> To solve the exact p-value using the hypergeometric distribution, only add probabilities of outcomes that have lower probabilities than the observed value in the direction of the alternative hypotheses.

## 4.4 Fisher's Exact Test: SAS

Including the `exact` or the `chisq` option under the `tables` statement will yield the results of the exact test.

```
proc freq;
weight count;
tables a*b/ exact chisq;
run;
```

## 4.5 Fisher's Exact Test: R

The `fisher.test()` can perform Fisher's exact test in `R` using a contingency table as input.

```
fisher.test(contingency_table,alternative="two.sided")
```

## 4.6 Example

The table shown shows data from a study on treatments for healing severe infections. Calculate the exact p-value of this contingency table using the Fisher's Exact Test.

- Under the alternative hypothesis: The test treatment is positively associated with the outcome.
- Under the alternative hypothesis: The test treatment is associated with the outcome.
- At a significance level of 0.10, do you have enough evidence to conclude that the treatment is associated with the outcome?

```
          Outcome
Treatment Favorable Unfavorable
  Test            5           1
  Control         1           3
```

## 4.7 Example

### 4.7.1 SAS Code

```
data severe;
input treat $ outcome $ count;
datalines;
Test f 5
Test u 1
Control f 1
Control u 3
;
proc freq order=data data=severe;
weight count;
tables treat*outcome / chisq nocol;
run;
```

### 4.7.2 R Code

```
dat <- expand.grid(Treatment=factor(c("Test","Control"),levels=c("Test","Control")),
                   Outcome = factor(c("Favorable","Unfavorable"),levels=c("Favorable","Unfavo
dat$count <- c(5,1,1,3)

dat_cc <- xtabs(count ~ Treatment+Outcome,data=dat)
dat_cc
```

```
         Outcome
Treatment Favorable Unfavorable
  Test            5           1
  Control         1           3
```

```
fisher.test(dat_cc, alternative = "greater") # one-sided
```

```
    Fisher's Exact Test for Count Data

data:  dat_cc
p-value = 0.119
alternative hypothesis: true odds ratio is greater than 1
95 percent confidence interval:
 0.5819    Inf
sample estimates:
odds ratio
  10.29391
```

```r
fisher.test(dat_cc,alternative= "two.sided")
```

```
    Fisher's Exact Test for Count Data

data:  dat_cc
p-value = 0.1905
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
   0.40161 930.24692
sample estimates:
odds ratio
  10.29391
```

### 4.7.3 Manual p-values: 1-sided

- One-sided hypothesis (+) corresponds to the the probabilities $P(X >= 5)$ which are equal to or less than $P(X = 5)$.

```r
N <- 5+1+1+3
m <- 6
k <- 6

dhyper(x=5,m=m,n=N-m,k=k)
```

```
[1] 0.1142857
```

For X $>$ 5, the only possible value of X is 6.

```r
dhyper(x=6,m=m,n=N-m,k=k)
```

```
[1] 0.004761905
```

The p-value is then the sum of these two probabilities:

```r
dhyper(x=5,m=m,n=N-m,k=k) + dhyper(x=6,m=m,n=N-m,k=k)
```

```
[1] 0.1190476
```

### 4.7.4 Manual p-values: 2-sided

- Two-sided hypothesis need to include values $P(X < 5)$ that have probabilities lower than $P(X = 5)$. These probabilities will be added to the one-sided p-value to get the two-sided p-value.

```
dhyper(x=4,m=m,n=N-m,k=k)
```

```
[1] 0.4285714
```

```
dhyper(x=3,m=m,n=N-m,k=k)
```

```
[1] 0.3809524
```

```
dhyper(x=2,m=m,n=N-m,k=k)
```

```
[1] 0.07142857
```

```
dhyper(x=1,m=m,n=N-m,k=k)
```

```
[1] 0
```

```
dhyper(x=0,m=m,n=N-m,k=k)
```

```
[1] 0
```

$P(X = 0)$, $P(X = 1)$, and $P(X = 2)$ have lower probabilities compared to $P(X = 5)$. Hence, we add these probabilities to the one-sided p-value to get the two-sided p-value.

```
dhyper(x=5,m=m,n=N-m,k=k) + dhyper(x=6,m=m,n=N-m,k=k) + dhyper(x=2,m=m,n=N-m,k=k) + dhyper(x=
```

```
[1] 0.1904762
```

### 4.7.5 Answers

- Under the alternative hypothesis: The test treatment is positively associated with the outcome. p=0.119
- Under the alternative hypothesis: The test treatment is associated with the outcome. p= 0.1905
- At a significance level of 0.10, do you have enough evidence to conclude that the treatment is associated with the outcome? **At $\alpha = 0.10$, we do not have sufficient evidence to conclude that treatment is associated with the outcomes in healing severe infections.**

> **Note**
>
> In SAS, be careful when selecting which one-sided alternative hypothesis to interpret from the results!

## 4.8 Exercise

Fisher's experiment: A number of individuals gathered together for afternoon tea. A lady in the group said that she could differentiate between whether the milk or the tea was poured first into a cup. Fisher suggested to pour tea first into four cups and to pour milk first into four cups. The lady was blinded to which cups had milk or tea poured first, but she knew there were four of each. The contingency table is shown below. At $\alpha = 0.05$, do we have sufficient evidence to conclude that the lady can differentiate tea based on how they were poured?

### 4.8.1 Contingency Table

```
      Guess
Actual Milk Tea
  Milk    3   1
  Tea     1   3
```

### 4.8.2 SAS Code

```
data tea;
input actual$ guess $ count;
datalines;
milk milk 3
milk tea 1
tea milk 1
```

```
tea tea 3
;

proc freq data=tea order=data;
weight count;
tables actual*guess/chisq;
run;
```

Direction is specified, hence we need to use the ">" one-sided p-value.

### 4.8.3 R `fisher.exact()`

Direction is specified, hence we need to use the ">" one-sided p-value.

```
dat <- expand.grid(Actual=factor(c("Milk","Tea"),levels=c("Milk","Tea")),
                   Guess = factor(c("Milk","Tea"),levels=c("Milk","Tea")))

dat$count <- c(3,1,1,3)

dat_cc <- xtabs(count ~ .,data=dat)
dat_cc
```

```
      Guess
Actual Milk Tea
  Milk    3   1
  Tea     1   3
```

```
fisher.test(dat_cc, alternative="greater")
```

```
    Fisher's Exact Test for Count Data

data:  dat_cc
p-value = 0.2429
alternative hypothesis: true odds ratio is greater than 1
95 percent confidence interval:
 0.3135693       Inf
sample estimates:
odds ratio
  6.408309
```

### 4.8.4 Hypergeometric: R

Direction is specified, hence we need to use the "$>$" one-sided p-value.

```
N <- 8
m <- 4
k <- 4

dhyper(x=3,m=m,n=N-m,k=k)
```

```
[1] 0.2285714
```

The only other possible value of $X$ is $X = 4$.

```
dhyper(x=4,m=m,n=N-m,k=k)
```

```
[1] 0.01428571
```

To get the one-sided p-value, we need to add these two probabilities:

```
dhyper(x=3,m=m,n=N-m,k=k) + dhyper(x=4,m=m,n=N-m,k=k)
```

```
[1] 0.2428571
```

> **Note**
>
> Note that we needed to add $P(X = 4)$ because its value is less than that of $P(X = 3)$.

### 4.8.5 Answer

The resulting p-value is 0.2429. At significance level $\alpha = 0.05$, we fail to reject the null hypothesis. We have no sufficient evidence to conclude that the lady can differentiate tea based on how they were poured.

# 5 Difference in Proportions

## 5.1 Difference in Proportions

We are often interested in comparing two independent populations. For binary responses, we can compare proportions of success.

> **Note**
>
> Recall from Intro to Stats course: we can estimate the confidence intervals and standardized statistics for the difference of two proportions.

## 5.2 Difference in Proportions: Estimation

Estimating the difference of the two proportions involves calculating the confidence intervals. There are different methods to calculate the confidence interval for the difference in proportion.

**Table 2.5** $2 \times 2$ Contingency Table

| | Yes | No | Total | Proportion Yes |
|---|---|---|---|---|
| Group 1 | $n_{11}$ | $n_{12}$ | $n_{1+}$ | $p_1 = n_{11}/n_{1+}$ |
| Group 2 | $n_{21}$ | $n_{22}$ | $n_{2+}$ | $p_2 = n_{21}/n_{2+}$ |
| Total | $n_{+1}$ | $n_{+2}$ | $n$ | |

> **Note**
>
> Independence of the two populations (Groups 1 and 2) mean that we have two independent binomial distributions.

## 5.3 Confidence Intervals: Wald

The commonly used confidence interval in estimating difference in proportions is the **Wald** confidence interval.

$$p_1 - p_2 \pm z_{1-\alpha/2}\sqrt{\frac{p_1(1-p_1)}{n_{1+}} + \frac{p_2(1-p_2)}{n_{2+}}}$$

> **Warning**
>
> This CI fails when $p_1$ or $p_2$ are close to the extreme values (0 and 1). this CI does not account for the discreteness of the data.

## 5.4 Confidence Intervals: Corrected Wald

We introduce a correction term to the Wald confidence interval to account for the continuity of the data.

$$p_1 - p_2 \pm z_{1-\alpha/2}\sqrt{\frac{p_1(1-p_1)}{n_{1+}} + \frac{p_2(1-p_2)}{n_{2+}}} + 1/2(1/n_{1+} + 1/n_{2+})$$

> **Warning**
>
> This CI still fails when $p_1$ or $p_2$ are close to the extreme values (0 and 1).

## 5.5 Confidence Intervals: Agresti-Caffo

We introduce a correction term to the Wald confidence interval to account for the extreme values.

$$p_1 - p_2 \pm z_{1-\alpha/2}\sqrt{\frac{\tilde{p}_1(1-\tilde{p}_1)}{n_{1+}+2} + \frac{\tilde{p}_2(1-\tilde{p}_2)}{n_{2+}+2}}$$

where

$$\tilde{p}_1 = (n_{11}+1)/(n_{1+}+2); \tilde{p}_2 = (n_{21}+1)/(n_{2+}+2)$$

> **Note**
>
> This correction keeps the coverage of the confidence interval at 95% for all possible values of $p_1$ and $p_2$.

## 5.6 Other Confidence Intervals

| Confidence Intervals | SAS code | R code | Adjustment Type |
|---|---|---|---|
| Wald | `wald` | "wald" | Default |
| Corrected Wald | `correct` | "waldcc" | Continuity correction |
| Agresti-Caffo | `ac` | "ac" | Improved Coverage |
| Miettinen-Nurminen | `mn` | "mn" | Uses goodness-of-fit to improve coverage |
| Newcombe | `newcombe` | "scorecc" | hybrid interval that does well for small counts |

| Confidence Intervals | SAS code | R code | Adjustment Type |
|---|---|---|---|
| Hauck-Anderson | `ha` | "ha" | Accounts for row totals |

## 5.7 Confidence Intervals: `SAS`

- The FREQ procedure has all mentioned confidence intervals
  - The options included in the `tables` statement are included in the previous slide.
  - The `riskdiff` option of the `tables` statement allows us to specify which confidence intervals we need for the difference as well as the estimates for the group proportions.
  - The `alpha` option can be used to change the confidence level.

Sample SAS code:

```
proc freq;
weight count;
tables treat*outcome/riskdiff(correct cl=(mn ac)) alpha=0.05;
run;
```

## 5.8 Confidence Intervals: `R`

The `BinomDiffCI()` function in the `DescTools` package in `R` can calculate these confidence intervals.

```
# install.packages("DescTools")
library(DescTools)
BinomDiffCI(x1=n11, n1 = row1sum, x2 = n21, n2=row2sum,
            conf.level=0.95,
            method = c("ac", "wald", "waldcc", "score", "scorecc", "mn",
                       "mee", "blj", "ha", "hal", "jp")
            )
```

## 5.9 Example

The contingency table shows the information from a randomized clinical trial that compared an experimental treatment to a placebo treatment for a respiratory disorder.

```
          Outcome
Treatment Favorable Unfavorable
  Placebo        16          48
  Test           40          20
```

### 5.9.1 Question

Calculate the 95% confidence interval for the difference in proportion of favorable outcomes for both treatment groups.

- Wald
- Corrected Wald
- Agresti-Caffo
- Miettinen-Nurminen

### 5.9.2 SAS Code

```
data respire;
input treat $ outcome $ count;
datalines;
test f 40
test u 20
placebo f 16
placebo u 48
;

proc freq data=respire order=data;
weight count;
tables treat*outcome/riskdiff(cl=(mn ac)) alpha=0.05;
tables treat*outcome/riskdiff(correct) alpha=0.05;
run;
```

### 5.9.3 R Code

```
dat <- expand.grid(Treatment=c("Placebo","Test"),
                   Outcome = c("Favorable","Unfavorable"))
dat$count <- c(16,40,48,20)

dat_cc <- xtabs(count ~ .,data=dat)
dat_cc
```

```
         Outcome
Treatment Favorable Unfavorable
  Placebo        16          48
  Test           40          20
```

```r
library(DescTools)
```

```
Warning: package 'DescTools' was built under R version 4.6.1
```

```r
BinomDiffCI(x1=40,
            n1 = 40+20,
            x2=16,
            n2=16+48,
            method=c('wald','waldcc','ac','mn'),
            conf.level = 0.95)
```

```
             est    lwr.ci    upr.ci
wald   0.4166667 0.2570362 0.5762972
waldcc 0.4166667 0.2408903 0.5924430
ac     0.4166667 0.2455745 0.5618546
mn     0.4166667 0.2460145 0.5626554
```

### 5.9.4 Answers

The estimated difference in proportion is 0.4167 with the following 95% confidence intervals: - Wald: **0.2570-0.5763** - Corrected Wald: **0.2409-0.5924** - Agresti-Caffo: **0.2456-0.5416** - Miettinen-Nurminen **0.2460-0.5627**

## 5.10 Exercise

One of the most famous and largest clinical trials ever performed was in 1954. The data obtained from this RCT is shown in the contingency table.

Salk vaccine clinical trial results; data source is Francis et al. (1955, p. 25).

| | Polio | Polio free | Total |
|---|---|---|---|
| Vaccine | 57 | 200,688 | 200,745 |
| Placebo | 142 | 201,087 | 201,229 |
| Total | 199 | 401,775 | 401,974 |

### 5.10.1 Problem

Calculate the *90%* confidence interval for the difference in proportion of polio-free children between the vaccine and placebo group.

- Corrected Wald
- Miettinen-Nurminen
- Newcombe

### 5.10.2 SAS Code

```
data vaxx;
length treat $7.;
input treat $ outcome $ count;
datalines;
vax n 200688
vax p 57
placebo n 201087
placebo p 142
;

proc freq data=vaxx order=data;
weight count;
tables treat*outcome/riskdiff(correct cl=(mn newcombe))
alpha=0.10;
run;
```

### 5.10.3 R Code

```
library(DescTools)
BinomDiffCI(x1=142,
            n1=201087+142,
            x2=57,
            n2 = 57+200688,
            method=c('waldcc','mn','newcombe'),
            conf.level = 0.9)
```

```
                est       lwr.ci       upr.ci
waldcc 0.0004217214 0.0003013909 0.0005420518
mn     0.0004217214 0.0003084844 0.0005405825
```

#### 5.10.4 Answer

The estimated difference in proportion is 0.0004.

- Corrected Wald **(0.0003,0.0005)**
- Miettinen-Nurminen **(0.0003,0.0005)**
- Newcombe **(0.0003,0.0005)**

# 6 Odds Ratio and Relative Risk

## 6.1 Odds Ratio (OR)

**Odds of success**

The ratio of probability of success and failure of an outcome is referred to as the odds.

$$Odds = p/(1-p)$$

**Odds Ratio**

The ratio of the odds of success for Group 1 and Group 2 is referred to as the odds ratio.

$$OR = \frac{p_1/(1-p_1)}{p_2/(1-p_2)} = (n_{11}n_{22}/n_{21}n_{12})$$

The odds ratio is commonly used as an effect size for retrospective studies. This statistic is also used in logistic regression techniques, which will be discussed later.

## 6.2 Relative Risk (RR)

The ratio of the risk developing a particular condition for individuals with risk factors present and absent is called the *relative risk*.

$$RR = p_1/p_2 = \frac{(n_{11}/n_{1+})}{(n_{21}/n_{2+})}$$

The relative risk is commonly used as an effect size in prospective studies.

> **Note**
>
> For cross-sectional data, the relative risk is known as the prevalence ratio because the disease and risk factor are assessed at the same time.

## 6.3 Odds Ratio & Relative Risk

- When OR, RR $> 1$, Group 1 is **more likely** than Group 2 to record success.
- When OR, RR$< 1$, Group 1 is **less likely** to record success.
- When OR, RR$= 1$, there is **no association** between group and success because the two groups are equally likely to record success.

> **Note**
>
> When the condition is rare, the OR is approximately equal to the RR.

## 6.4 Odds Ratio & Relative Risk: `SAS`

- OR and RR are calculated using the measures option in the TABLES statement in PROC FREQ.
- For small sample sizes, the exact confidence limits can be calculated for the odds ratios.

```
proc freq;
weight count;
tables a*b/measures;
exact or rr;
run;
```

## 6.5 Odds Ratio & Relative Risk: `R`

The odds ratio and relative risk can be calculated using functions from the `epitools` package.

```
library(epitools)
epitools::oddsratio(contingency_table,
          conf.level = 0.95,
          method=c("fisher","wald"),
          rev=c("rows","columns","both")
)
```

```
epitools::riskratio(contingency_table,
          conf.level = 0.95,
          method=c("fisher","wald"),
          rev=c("rows","columns","both")
)
```

> **Tip**
>
> The `fisher` method provides exact confidence intervals for relative risk and odds ratio. The `fisher.test()` function provides an exact confidence interval for the odds ratio based on the hypergeometric distribution, but it does not include the relative risk.

> **Tip**
>
> The `rev` argument allows us to re-arrange the contingency table without defining factors to establish row and column `R`. The order of the columns for `oddsratio` and `riskratio` should follow this order:
>
> - Rows (top to bottom): No exposure (risk absent), Exposure (risk present)
> - Cols (left to right): Control (Outcome=0), Case (Outcome=1)

## 6.6 Example

The data set `severe_long.csv` includes data from a study on treatments for severe infections.

### 6.6.1 Problem

Calculate the asymptotic and exact confidence limits for the odds ratio of favorable outcomes for test and control groups.

```
dat <- read.csv("severe_long.csv")
head(dat)
```

```
  Treatment   Outcome
1      Test Favorable
2      Test Favorable
3      Test Favorable
4      Test Favorable
5      Test Favorable
6      Test Favorable
```

### 6.6.2 SAS Code

Import severe_long as work.severe_long in SAS.

```
proc freq order=data data=work.severe_long;
tables treat*outcome;
exact or;
run;
```

| Method | Value and CI |
|---|---|
| Asymptotic | 10 (1.0256, 97.5005) |
| Exact | 10 (0.6896,166.3562) |

**The two methods lead to different conclusions.**

### 6.6.3 R Code

Create the contingency table.

```
dat_cc <- xtabs(~Treatment+Outcome,data=dat)
dat_cc
```

```
         Outcome
Treatment Favorable Unfavorable
  Control         2           4
  Test           10           2
```

Calculate the odds ratio and the confidence intervals.

```
library(epitools)
```

```
Warning: package 'epitools' was built under R version 4.6.1
```

```
Attaching package: 'epitools'
```

```
The following object is masked from 'package:vcdExtra':

    expand.table
```

```
The following object is masked from 'package:vcd':

    oddsratio
```

```
epitools::oddsratio(dat_cc,
          method=c("wald"),
          conf.level=0.95,
          rev="columns" #Reverses the columns
          )
```

```
Warning in chisq.test(xx, correct = correction): Chi-squared approximation may
be incorrect
```

```
$data
         Outcome
Treatment Unfavorable Favorable Total
  Control           4         2     6
  Test              2        10    12
  Total             6        12    18

$measure
         odds ratio with 95% C.I.
Treatment estimate    lower    upper
  Control        1       NA       NA
  Test          10 1.025636 97.50049

$p.value
         two-sided
Treatment midp.exact fisher.exact chi.square
  Control         NA           NA         NA
  Test    0.06119371    0.1070351 0.03389485

$correction
[1] FALSE

attr(,"method")
[1] "Unconditional MLE & normal approximation (Wald) CI"
```

```
epitools::oddsratio(dat_cc,
          method=c("fisher"),
          conf.level=0.95,
```

```
          rev="columns" #Reverses the columns,
          )
```

```
Warning in chisq.test(xx, correct = correction): Chi-squared approximation may
be incorrect
```

```
$data
         Outcome
Treatment Unfavorable Favorable Total
  Control           4         2     6
  Test              2        10    12
  Total             6        12    18

$measure
         odds ratio with 95% C.I.
Treatment estimate     lower    upper
  Control 1.000000        NA       NA
  Test    8.457238 0.6896384 166.4345

$p.value
         two-sided
Treatment midp.exact fisher.exact chi.square
  Control         NA           NA         NA
  Test    0.06119371    0.1070351 0.03389485

$correction
[1] FALSE

attr(,"method")
[1] "Conditional MLE & exact CI from 'fisher.test'"
```

> **Note**
>
> We report the **estimate** of the odds ratio from the `"wald"` method, but use the `"fisher"` confidence interval as the exact confidence interval.
> We also use `epitools::oddsratio` to tell `R` that we are using the `oddsratio` function in `epitools` and not another package (`vcd` also has a package of the same name).

## 6.7 Exercise

Two groups having the same symptoms of respiratory illness were sampled. One group was treated with an experimental treatment and the other group was treated with a placebo. The participants were asked whether there was an overall treatment in overall respiration.

### 6.7.1 Data

```
          Outcome
Treatment Yes No
  Test     29 16
  Placebo  14 31
```

### 6.7.2 SAS Code

```
data resp;
input  Treatment $ Outcome  $ count;
datalines;
Test Yes 29
Placebo Yes 14
Test No 16
Placebo No 31
;
proc freq data=resp order=data;
weight count;
tables treatment*outcome/measures;
run;
```

### 6.7.3 R Code

```
dat <- expand.grid(Treatment=c("Test","Placebo"),
                   Outcome = c("Yes","No"))
dat$count = c(29,14,16,31)

dat_cc<- xtabs(count ~ .,data=dat)
dat_cc
```

```
          Outcome
Treatment Yes No
```

```
  Test     29 16
  Placebo  14 31
```

Odds ratio

```
library(epitools)
or <- epitools::oddsratio(dat_cc,
                          method="wald",
                          conf.level=0.95,
                          rev="both")
or
```

```
$data
         Outcome
Treatment No Yes Total
  Placebo 31  14    45
  Test    16  29    45
  Total   47  43    90

$measure
         odds ratio with 95% C.I.
Treatment estimate    lower    upper
  Placebo 1.000000       NA       NA
  Test    4.013393 1.668039 9.656441

$p.value
         two-sided
Treatment  midp.exact fisher.exact  chi.square
  Placebo          NA           NA          NA
  Test    0.001783016  0.002917802 0.001548669

$correction
[1] FALSE

attr(,"method")
[1] "Unconditional MLE & normal approximation (Wald) CI"
```

```
or$measure
```

```
         odds ratio with 95% C.I.
Treatment estimate    lower    upper
```

```
  Placebo 1.000000       NA       NA
  Test    4.013393 1.668039 9.656441
```

Relative Risk

```
rr <- riskratio(dat_cc,
                method="wald",
                rev="both")
rr
```

```
$data
         Outcome
Treatment No Yes Total
  Placebo 31  14    45
  Test    16  29    45
  Total   47  43    90

$measure
         risk ratio with 95% C.I.
Treatment estimate    lower    upper
  Placebo 1.000000       NA       NA
  Test    2.071429 1.274194 3.367474

$p.value
         two-sided
Treatment  midp.exact fisher.exact  chi.square
  Placebo          NA           NA          NA
  Test    0.001783016  0.002917802 0.001548669

$correction
[1] FALSE

attr(,"method")
[1] "Unconditional MLE & normal approximation (Wald) CI"
```

```
rr$measure
```

```
         risk ratio with 95% C.I.
Treatment estimate    lower    upper
  Placebo 1.000000       NA       NA
  Test    2.071429 1.274194 3.367474
```

#### 6.7.4 Answer

- Calculate the odds ratio of respiratory improvement between test and placebo groups.**4.01 (1.67-9.66)**
- What is the relative risk of having respiratory improvement if we consider the treatment as a risk factor? **2.07 (1.27-3.37)**

# 7 Sensitivity and specificity

## 7.1 Sensitivity

True proportion of positive results that a test elicits when performed on subjects known to have the disease.

$$Sens = TP/(TP + FN) = n_{11}/n_{1+}$$

Sensitivity is the conditional probability of a positive test given that the subject has the disease ($P(PT|HD)$)

## 7.2 Specificity

True proportion of negative results that a test elicits when performed on subjects known to be disease free.

$$Sens = TN/(TN + FP) = n_{22}/n_{2+}$$

Specificity is the conditional probability of a negative test when the subject does not have the disease ($P(NT|ND)$).

## 7.3 Sensitivity and Specificity: SAS

The `FREQ` procedure can also calculate sensitivity and specificity.

- `riskdiff` approximates these metrics using the risk estimates.
  - Sensitivity: Row 1 risk estimate for Column 1
  - Specificity: Row 2 risk estimate for Column 2

## 7.4 Sensitivity and Specificity: R

The epiR package has the epi.tests() function that calculates sensitivity and specificity. For the following contingency table:

| | Outcome + | Outcome - |
|---|---|---|
| Test + | $n_{11}$ | $n_{12}$ |
| Test - | $n_{21}$ | $n_{22}$ |

```
install.packages("epiR")
library(epiR)

epi.tests(contingency_table,
          method=c("exact","wilson","agresti","clopper-pearson","jeffreys"))
```

> **Note**
>
> The default method used to calculate confidence intervals for the sensitivity and specificity is the exact method.

## 7.5 Example

The following table shows the results of a study investigating a new screening device for a skin disease. Calculate the sensitivity and specificity of the screening device with the corresponding exact 95% confidence interval.

### 7.5.1 Data

```
          Status
Test       Disease Present Disease Absent
  Positive              52             20
  Negative               8            100
```

### 7.5.2 SAS Code

```
data screening;
input disease $ test $ count @@;
datalines;
present + 52 present - 8
absent + 20 absent - 100
;
proc freq data=screening order=data;
weight count;
tables disease*test / riskdiff;
run;
```

### 7.5.3 R Code

```
dat <- expand.grid(Status=c("Disease Present","Disease Absent"),
                   Test = c("Positive","Negative")) |>
mutate(count = c(52,20,8,100))

xtabs(count ~ Test+Status,data=dat) -> dat_cc
dat_cc
```

```
          Status
Test       Disease Present Disease Absent
  Positive              52             20
  Negative               8            100
```

```
library(epiR)
```

```
Warning: package 'epiR' was built under R version 4.6.1
```

```
Loading required package: survival
```

```
Attaching package: 'survival'
```

```
The following object is masked from 'package:epitools':

    ratetable
```

```
Package epiR 2.0.96 is loaded
```

```
Type help(epi.about) for summary information

Type browseVignettes(package = 'epiR') to learn how to use epiR for applied epidemiological a
```

```
epi.tests(dat_cc)
```

```
          Outcome +    Outcome -      Total
Test +           52           20         72
Test -            8          100        108
Total            60          120        180

Point estimates and 95% CIs:
-------------------------------------------------------------
Apparent prevalence *                  0.40 (0.33, 0.48)
True prevalence *                      0.33 (0.26, 0.41)
Sensitivity *                          0.87 (0.75, 0.94)
Specificity *                          0.83 (0.75, 0.90)
Positive predictive value *            0.72 (0.60, 0.82)
Negative predictive value *            0.93 (0.86, 0.97)
Positive likelihood ratio              5.20 (3.44, 7.85)
Negative likelihood ratio              0.16 (0.08, 0.31)
False T+ proportion for true D- *      0.17 (0.10, 0.25)
False T- proportion for true D+ *      0.13 (0.06, 0.25)
False T+ proportion for T+ *           0.28 (0.18, 0.40)
False T- proportion for T- *           0.07 (0.03, 0.14)
Correctly classified proportion *      0.84 (0.78, 0.89)
-------------------------------------------------------------
* Exact CIs
```

### 7.5.4 Answer

- Sensitivity: 0.87 (0.75-0.94)
- Specificity: 0.83 (0.75-0.90)

## 7.6 Exercise

A pharmaceutical company developed a screening test that can be used to detect infection of the Lassa virus. The newly developed test was said to be cheaper than the tests currently available to the public, but its performance was yet to be evaluated. The screening test was applied to 115 diagnosed cases (w/ Lassa virus) and 120 control participants (w/o Lassa virus) of the Lassa fever. The resulting contingency table is shown in the table.

### 7.6.1 Question

- What is the sensitivity of the test?
- What is the specificity of the test?

| | Test | |
|---|---|---|
| Status | Positive | Negative |
| Disease Present | 97 | 18 |
| Disease Absent | 45 | 75 |

### 7.6.2 R Code

```
library(tidyverse)
dat <- expand.grid(Status=c("Disease Present","Disease Absent"),
                   Test = c("Positive","Negative"))
dat$count = c(97,45,18,75)

dat_cc <- xtabs(count ~ Test+Status,data=dat)
dat_cc
```

```
          Status
Test       Disease Present Disease Absent
  Positive              97             45
  Negative              18             75
```

```
library(epiR)
epi.tests(dat_cc)
```

```
          Outcome +    Outcome -      Total
Test +           97           45        142
Test -           18           75         93
```

```
Total           115          120        235

Point estimates and 95% CIs:
-------------------------------------------------------------
Apparent prevalence *                  0.60 (0.54, 0.67)
True prevalence *                      0.49 (0.42, 0.56)
Sensitivity *                          0.84 (0.76, 0.90)
Specificity *                          0.62 (0.53, 0.71)
Positive predictive value *            0.68 (0.60, 0.76)
Negative predictive value *            0.81 (0.71, 0.88)
Positive likelihood ratio              2.25 (1.76, 2.87)
Negative likelihood ratio              0.25 (0.16, 0.39)
False T+ proportion for true D- *      0.38 (0.29, 0.47)
False T- proportion for true D+ *      0.16 (0.10, 0.24)
False T+ proportion for T+ *           0.32 (0.24, 0.40)
False T- proportion for T- *           0.19 (0.12, 0.29)
Correctly classified proportion *      0.73 (0.67, 0.79)
-------------------------------------------------------------
* Exact CIs
```

### 7.6.3 SAS Code

```
data screening;
input disease $ outcome $ count @@;
datalines;
present + 97 present - 18
absent + 45 absent - 75
;
proc freq data=screening order=data;
weight count;
tables disease*outcome / riskdiff;
run;
```

### 7.6.4 Answer

- What is the sensitivity of the test? **0.8435 (0.76, 0.90)**
- What is the specificity of the test? **0.6250 (0.53, 0.71)**

# 8 McNemar's Test

## 8.1 McNemar's Test

When the 2x2 table provides information from matched pairs, the two responses are related to each other. Hence, independence cannot be assumed.

**Null Hypothesis**

$H_0$: The marginal probabilities of each outcome are the same.

**Test Statistic**

The test statistic for the McNemar's test is $Q_M$ given by

$$Q_M = (n_{12} - n_{21})^2/(n_{12} + n_{21})$$

## 8.2 McNemar's Test: `SAS`

- The `agree` option in the `tables` statement can be used to perform the McNemar's test.
- Exact p-values can be calculated using the `exact` statement.

### 8.2.1 Sample `SAS` Code

```
proc freq;
weight count;
tables a*b/agree;
exact mcnem;
run;
```

## 8.3 McNemar's Test: `R`

The `mcnemar.test()` function can be used to perform the McNemar's test.

```
mcnemar.test(contingency_table) # no continuity correction
```

> **Note**
>
> The p-value calculated using this function adds the continuity correction by default.

## 8.4 Example

The table displays data collected by political science researchers who polled husbands and wives on whether they approved of one of their U.S. senators. The cell counts represent the number of pairs of husbands and wives who fit the configurations indicated by the row and column levels. Use the McNemar's test to determine whether there is no difference in husband and wife approvals on the state senator.

### 8.4.1 Data

```
       Wife
Husband Yes No
    Yes  20  5
    No   10 10
```

### 8.4.2 SAS Code

```
data approval;
input hus_resp $ wif_resp $ count;
datalines;
yes yes 20
yes no 5
no yes 10
no no 10
;

proc freq order=data;
weight count;
tables hus_resp*wif_resp / agree;
exact mcnem;
run;
```

### 8.4.3 R Code

```
dat <- expand.grid(Husband=c("Yes","No"),
                   Wife = c("Yes","No"))
dat$count <- c(20,10,5,10)

dat_cc <- xtabs(count ~ Husband+Wife,data=dat)
dat_cc
```

```
       Wife
Husband Yes No
    Yes  20  5
    No   10 10
```

```
mcnemar.test(dat_cc)
```

```

    McNemar's Chi-squared test with continuity correction

data:  dat_cc
McNemar's chi-squared = 1.0667, df = 1, p-value = 0.3017
```

> **Note**
>
> If we want to recover the asymptotic p-value from SAS, we need to remove the continuity correction.
>
> ```
> mcnemar.test(dat_cc,correct=F)
> ```
>
> ```
>
>     McNemar's Chi-squared test
>
> data:  dat_cc
> McNemar's chi-squared = 1.6667, df = 1, p-value = 0.1967
> ```

### 8.4.4 Answer

The exact/corrected p-value = 0.3018, which implies that there is no evidence of a difference between the senator approval ratings of the husbands and wives.

## 8.5 Exercise

The tables show the data from a study that compared the diagnostic accuracy of magnetic resonance imaging (MRI) and ultrasound in patients who had been established as having localized prostate cancer by a gold standard test.

### 8.5.1 Question

Use the McNemar's test to determine whether the two methods have a different probabilities in diagnosing localized prostate cancer. (*Hint: Small cell counts*)

```
           Ultrasound
MRI         Localized Advanced
  Localized         4        6
  Advanced          3        3
```

### 8.5.2 R Code

```
           Ultrasound
MRI         Localized Advanced
  Localized         4        6
  Advanced          3        3
```

```
mcnemar.test(dat_cc)
```

```

    McNemar's Chi-squared test with continuity correction

data:  dat_cc
McNemar's chi-squared = 0.44444, df = 1, p-value = 0.505
```

### 8.5.3 SAS Code

```
data imaging;
input mri $ ult $ count;
datalines;
loc loc 4
loc adv 6
adv loc 3
adv adv 3
```

```
;

proc freq order=data;
weight count;
tables mri*ult / agree;
exact mcnem;
run;
```

#### 8.5.4 Answer

Use the McNemar's test to determine whether the two methods have a different probabilities in diagnosing localized prostate cancer. (*Hint: Small cell counts*)

- Use exact p-value
- $p = 0.51$. No evidence of any difference in probabilities in correctly diagnosing localized prostate cancer.

# 9 Incidence Densities

## 9.1 Incidence Densities

- Counts of subjects who responded within a duration of exposure to an event.
- Poisson distribution: counts
- Examples:
  - Bacteria colony count
  - Disease incidence
  - Accidents
  - Vaccine failure
- Common measure of duration of exposure/ time-at-risk: person-time

## 9.2 Poisson Distribution

Recall that the Poisson distribution is described by the following density function:

$$P(x) = e^{-\lambda}\lambda^x/x!$$

where $\lambda$ is the expected number of events at a given unit of time.

- is also referred to as incidence density.
- For $t$ units of time, the expected number of events scales to $\lambda t$.
- Main assumption: The time to event is a random variable following an exponential distribution.

## 9.3 Incidence Density Ratios

- The table shows the number of infections (events) and person years for participants of an RCT.
  - $n_v$ and $n_p$ are Poisson random variables with respective incidence densities equal to $\lambda_v$ and $\lambda_p$
- If $\lambda_v$ is the incidence density for the vaccine group, then the expected number of events is $\lambda_v N_v$.
- Similarly, the expected number of events in the placebo group is $\lambda_p N_p$
- We are interested in testing the null hypothesis $\lambda_v/\lambda_p = 1$.
  - $\lambda_v/\lambda_p$ is referred to as the **Incidence Density Ratio (IDR)**

**Table 2.14** Vaccine Study

| Treatment | Events | Person Years |
|---|---|---|
| Vaccine | $n_v$ | $N_v$ |
| Placebo | $n_p$ | $N_p$ |

## 9.4 Incidence Density Ratios

The IDR can be estimated using conditional binomial distributions assuming $n_v$ and $n_p$ are independent Poisson processes.

$$P(n_v|n_v + n_p) \sim Bin\left(N = n_v + n_p, p = \frac{\lambda_v N_v}{\lambda_v N_v + \lambda_p N_p}\right)$$

## 9.5 Incidence Density Ratios: Confidence Interval

The binomial proportion can be estimated using the proportion of events in the treatment group:

$$p = \frac{n_v}{n_v + n_p} = \frac{IDR(N_v/N_p)}{1 + (IDR)(N_v/N_p)}$$

Using algebra (Problem Set 1), *it is easy to show that*:

$$IDR = \frac{p}{1-p}\left(N_p/N_v\right)$$

## 9.6 Incidence Density Ratios: Confidence Interval

You can calculate the confidence interval estimates for $p$ using the binomial distribution assumption $(p_L, p_U)$ and apply the scaling factor to each confidence limit.

$$95\%CI : \left(\frac{p_L}{1-p_L}(N_p/N_v), \frac{p_U}{1-p_U}(N_p/N_v)\right)$$

## 9.7 Incidence Density Ratios: `SAS`

- The `binomial` option in the `tables` statement produces the estimate and exact confidence limits for the binomial proportion $p$, i.e. $p_L$ and $p_U$.
- We use the estimate and exact confidence limits of $p$ in calculating the IDR.
- The `IML` procedure, which is the procedure used to perform mathematical equations, is used to calculate the IDR confidence limits.
  - Later on, we will use the `GENMOD` procedure to calculate exact confidence intervals for the IDR.

## 9.8 Incidence Density Ratios: `R`

The `epitools` package includes the `rateratio()` function that can calculate an approximation of the incidence density ratio.

```
rateratio(contingency_table)
#OR

rateratio(x=events,y=person_years)
```

## 9.9 Example

The table shows the results of an international study of the effect of a rotavirus vaccine in nearly 70,000 infants on health care events. The events counted were hospitalizations or emergency room visits.

### 9.9.1 Questions

- Estimate the IDR of hospitalization/emergency room visits between the two groups.
- Calculate the 95% CI.

**Table 2.13** Vaccine Study Results for U.S.

| Treatment | Events | Person Years |
|---|---|---|
| Vaccine | 3 | 7500 |
| Placebo | 58 | 7250 |

### 9.9.2 SAS Code

```
data vaccine2;
input Outcome $ Count @@;
datalines;
vaccine 3 placebo 58
;

ods select BinomialCLs; *this statement filters the output.;


proc freq order=data data=vaccine2;                                    ①
weight count;
tables Outcome / binomial (exact); *calculates the exact confidence limits for the binomial p
ods output BinomialCLs=BinomialCLs; *outputs a dataset that includes the binomial CLs.;
run;
```

| IDR | IDR_lower | IDR_higher |
|---|---|---|
| 0.05 | 0.01002014 | 0.1535468 |

```

proc iml ;
Use BinomialCLs var{LowerCL UpperCL};
read all into CL;
print CL;
q = { 3, 7500, 58 , 7250 }; *vector of numbers from the table;
C= q[2]/q[4]; *q[i] represents the ith number in the q vector defined above. C is the person
P= q[1]/ (q[1] + q[3]); *estimate of the binomial distribution;
R= P/ ((1-P) *C) ; * IDR estimate;
CI= CL[1]/((1-CL[1])*C) || CL[2]/((1-CL[2])*C) ; *IDR CL;
print r CI;
quit;
```

① `order=data` is important so as to keep the proportions consistent.

### 9.9.3 R Code

```
events <- c(58,3)
person_years <- c(7250,7500)

mult <- person_years[1]/person_years[2] #multiplier
DescTools::BinomCI(events[2],sum(events),method = "clopper-pearson") -> p_est



data.frame(IDR = p_est[1]/(1-p_est[1])*mult,
           IDR_lower = p_est[2]/(1-p_est[2])*mult,
          IDR_higher = p_est[3]/(1-p_est[3])*mult) |> gt::gt()
```

```
rateratio(x=events,y=person_years)
```

```
$data
          Outcome
Predictor  events person_years
  Exposed1     58         7250
```

```
  Exposed2      3         7500
  Total        61        14750

$measure
          rate ratio with 95% C.I.
Predictor    estimate      lower     upper
  Exposed1 1.00000000         NA        NA
  Exposed2 0.05253473 0.01239861 0.1418922

$p.value
          two-sided
Predictor               midp.exact                   wald
  Exposed1                      NA                     NA
  Exposed2 0.00000000000006708884 0.0000000000007207568

attr(,"method")
[1] "Median unbiased estimate & mid-p exact CI"
```

### 9.9.4 Answers

- Estimate the IDR of hospitalization/emergency room visits between the two groups. **IDR = 0.05**
- Calculate the 95% CI. **(0.01-0.15)** or with `riskratio`: (0.01-0.14)

## 9.10 Exercise

The table lists the number of accidents for day and night shift workers at a certain factory in Southeast Asia. The company wants you to investigate if there is any evidence of night shift workers having a higher accident density, i.e., higher incidence density of accidents, compared to the day shift.

### 9.10.1 Data

```
       Metric
Shift   Accidents Person-Hours
  Night        25         1150
  Day          20         1240
```

### 9.10.2 SAS Code

```
data accident;
/*data only has the events. Person-Hours are used later.*/
input Outcome $ Count @@;
datalines;
night 25 day 20                                                                    ①
;

ods select BinomialCLs; *this statement filters the output.;

proc freq order=data data=accident;
weight count;
tables Outcome / binomial (exact) alpha=0.05; *calculates the exact confidence limits for the
ods output BinomialCLs=BinomialCLs; *outputs a dataset that includes the binomial CLs.;
run;

proc iml ;
Use BinomialCLs var{LowerCL UpperCL};
read all into CL;
print CL;
q = { 25 1150 20 1240 }; *vector of numbers from the table;
C= q[2]/q[4]; *q[i] represents the ith number in the q vector defined above. C is the person-
P= q[1]/ (q[1] + q[3]); *estimate of the binomial distribution;
R= P/ ((1-P) *C) ; * IDR estimate;
CI= CL[1]/((1-CL[1])*C) || CL[2]/((1-CL[2])*C) ; *IDR CL;
print r CI;
quit;
```

① "Success" is an event happening during the night shift.

### 9.10.3 R Code

```
events <- c(20,25)
person_years <- c(1240,1150)

mult <- person_years[1]/person_years[2] #multiplier
DescTools::BinomCI(events[2],sum(events),method = "clopper-pearson") -> p_est
```

| IDR | IDR_lower | IDR_higher |
|---|---|---|
| 1.347826 | 0.7187612 | 2.559057 |

```
data.frame(IDR = p_est[1]/(1-p_est[1])*mult,
           IDR_lower = p_est[2]/(1-p_est[2])*mult,
          IDR_higher = p_est[3]/(1-p_est[3])*mult) |> gt::gt()
```

Using `rateratio()`

```
rateratio(x=events,y=person_years)
```

```
$data
          Outcome
Predictor  events person_years
  Exposed1     20         1240
  Exposed2     25         1150
  Total        45         2390

$measure
          rate ratio with 95% C.I.
Predictor  estimate     lower   upper
  Exposed1 1.000000        NA      NA
  Exposed2 1.345627 0.7466559 2.45738

$p.value
          two-sided
Predictor  midp.exact      wald
  Exposed1         NA        NA
  Exposed2  0.3232959 0.3179524

attr(,"method")
[1] "Median unbiased estimate & mid-p exact CI"
```

### 9.10.4 Answer

- IDR: 1.35 (0.72-2.56)
- with `riskratio`: (0.75-2.46)

# 10 Power Analysis

## 10.1 Power Analysis

- Power is defined as the probability of rejecting the null hypothesis when it is false.
  - Acceptable range: 80% and above.
- Also known as (a priori) Sample Size Calculations
  - How many samples do I need to detect a set difference between the treatment and placebo group at a significance level of 0.05?

## 10.2 Power Analysis: SAS

- The `POWER` procedure enables us to calculate how many samples are needed to detect a set effect size in examining the difference between two proportions from two different populations.
- Specifically, the TWOSAMPLEFREQ option
  - RCT: treatment vs. control
- We need to set an effect size (theory-based), significance level, and power level.
  - Other options can be found here

## 10.3 Example

Suppose prior literature states that the incidence rate of a chronic disease is 0.40. Suppose we want to perform a prospective study to test the association between the disease incidence and family history, i.e. whether other family members were diagnosed with the chronic disease. We want to detect a proportion difference of 0.10 between groups with power of 80% (0.80) at a significance level of 0.05.

## 10.4 Example

Suppose prior literature states that the incidence rate of a chronic disease is 0.40. Suppose we want to perform a prospective study to test the association between the disease incidence and family history, i.e. whether other family members were diagnosed with the chronic disease. We want to detect a proportion difference of 0.10 between groups with power of 80% (0.80) at a significance level of 0.05.

- We need 388 units per group (family +, family -). This brings us to 776 total units.

### 10.4.1 SAS Code

```
proc power ;
twosamplefreq
    refproportion=0.4
    alpha=0.05
    power=0.80
    proportiondiff=0.1
    npergroup=.;

run;
```

### 10.4.2 R Code

```
power.prop.test(p1=0.4,p2=0.5,power=0.8)
```

```

     Two-sample comparison of proportions power calculation

              n = 387.3385
             p1 = 0.4
             p2 = 0.5
      sig.level = 0.05
          power = 0.8
    alternative = two.sided

NOTE: n is number in *each* group
```